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INTERNSHIP PROJECT REPORT

CUSTOMER SEGMENTATION & CHRUN PATTERN ANALYTICS IN EUROPEAN BANKING

A Finance Analyst project

Detailed Project Report

Submitted as part of the Unified Mentor project

PREPARED BY:

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PROJECT DOMAIN

Finance Analyst

TOOLS USED

Microsoft Excel | Python | Streamlit

SUBMISSION YEAR

2026

DECLARATION

I hereby declare that the project report titled **"Customer Segmentation & Churn Pattern Analytics in European Banking"** is my original work completed as part of my internship with **Unified Mentor** in the domain of **Finance Analyst**.

The analysis presented in this report is based on the European banking customer dataset and has been carried out for academic and learning purposes. I have completed the analysis, visualization, and dashboard development as part of the assigned internship project.

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PROJECT DOMAIN: Finance Analyst
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Year: 2026

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to **Unified Mentor** for providing me with the opportunity to undertake this internship project on **"Customer Segmentation & Churn Pattern Analytics in European Banking."**

I am grateful for the guidance and learning opportunities provided throughout the project. This internship helped me develop practical knowledge in financial analysis, customer segmentation, data analysis, visualization, and dashboard development.

I would also like to thank everyone who supported and encouraged me during the completion of this project. The experience has enhanced my analytical skills and provided valuable exposure to applying data-driven approaches to business and finance-related problems.

Finally, I am thankful for the opportunity to work on this project and gain practical experience as a **Finance Analyst intern**.

MAKANE VAISHNAVI

FINANCE ANALYST INTERN

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1. ABSTRACT

This project focuses on analyzing customer segmentation and churn patterns in the European banking customer dataset. The objective of the study is to identify customer segments with different levels of observed churn and generate insights that can support data-driven customer retention strategies.

The analysis was performed using **Microsoft Excel, Python, and Streamlit**. The dataset contains information on **10,000 customers**, including demographic, financial, product usage, geographic, and engagement-related attributes. The target variable used for churn analysis was **Exited**, which indicates whether a customer left the bank.

The overall observed churn rate in the dataset was **20.37%**. The analysis showed notable differences across customer segments. Germany recorded an observed churn rate of **32.44%**, while customers aged **46–60 years** showed an observed churn rate of **51.12%**. In terms of engagement, inactive customers showed a churn rate of **26.85%**, compared with **14.27%** among active customers. Female customers showed a churn rate of **25.07%**, compared with **16.46%** for male customers.

A Streamlit dashboard was developed to provide interactive analysis through filters, key performance indicators, visualizations, and a customer-level explorer. The project demonstrates how structured data analysis and visualization can be used to identify important customer segments and support evidence-based business analysis in the banking sector.

2. INTRODUCTION

Customer retention is an important area of analysis in the banking sector because customers interact with financial institutions through multiple products, services, and channels. Understanding customer behaviour can help organizations identify customer segments that show different patterns of retention and churn.

Customer churn refers to customers who leave or discontinue their relationship with an organization. In a banking context, analysing churn can provide useful information about differences between customer groups based on factors such as age, geography, gender, account balance, credit score, product usage, and customer engagement.

This project analyses a **European banking customer dataset containing 10,000 customer records**. The analysis examines different customer characteristics and compares their observed churn rates. The project uses **Microsoft Excel** for data preparation and exploratory analysis, **Python** for analytical processing, and **Streamlit** for developing an interactive dashboard.

The analysis identified several differences in observed churn across customer segments. For example, Germany recorded an observed churn rate of **32.44%**, while the 46–60 age group recorded **51.12%**. Inactive customers also showed a higher observed churn rate of **26.85%** compared with **14.27%** among active customers.

The purpose of this project is not only to calculate churn rates but also to present the findings in an accessible and interactive format. The Streamlit dashboard allows users to apply filters, view dynamic key performance indicators, examine visualizations, and explore individual customer records. This provides a practical demonstration of how data analysis can support customer-focused business and financial decision-making.

3. PROBLEM STATENMENT

Customer churn is an important business concern for banking institutions because the loss of customers can affect customer relationships, product usage, and overall business performance. Identifying customer segments with different levels of observed churn can help organizations understand where greater attention may be required.

The European banking customer dataset contains information about customers across different geographic, demographic, financial, product, and engagement characteristics. However, analysing these variables individually can make it difficult to identify meaningful patterns in customer churn.

Therefore, this project focuses on analysing customer data to identify differences in observed churn across various customer segments, including geography, age, gender, tenure, credit score, number of products, balance, salary, and customer engagement.

The project also aims to present these findings through an interactive **Streamlit dashboard**, allowing users to apply filters, view dynamic key performance indicators, analyse charts, and explore customer-level information.

The problem addressed by this project is therefore to **identify and communicate customer churn patterns through systematic data analysis and interactive visualization**, so that the resulting insights can support data-driven business analysis and customer retention planning.

4. PROJECT OBJECTIVES

4.1 Primary Objective

The primary objective of this project is to analyse customer churn patterns in the European banking customer dataset using data analysis and visualization techniques, and to present actionable insights through an interactive Streamlit dashboard.

4.2 Specific Objectives

The specific objectives of the project are:

1. To analyse customer data and understand the overall churn rate.
2. To examine churn patterns across geographical, demographic, financial, and customer engagement segments.
3. To calculate and compare churn rates based on age, gender, geography, tenure, credit score, number of products, balance, and salary.
4. To identify customer segments that show relatively higher observed churn rates.
5. To develop an interactive dashboard using Streamlit to present key performance indicators, charts, filters, and customer-level information.
6. To communicate the findings through meaningful visualizations and business recommendations.
7. To demonstrate the practical application of Microsoft Excel, Python, and Streamlit in data analysis.

5. DATASET DESCRIPTION

The project uses a European banking customer churn dataset containing 10,000 customer records. It includes demographic, geographic, financial, product-related, and customer engagement information.

The analysis uses variables such as CreditScore, Geography, Gender, Age, Tenure, Balance, Number Of Products, Is Active Member, Estimated Salary, and Exited. The Exited variable identifies whether a customer left the bank (1) or remained (0).

The dataset supports customer segmentation and comparison of churn rates across different customer groups. It was analysed using Microsoft Excel and Python, with the findings presented through an interactive Streamlit dashboard.

6. ANALYTICAL METHODOLOGY

The project follows a systematic approach to analyse customer churn patterns in the European banking customer dataset. The methodology includes data review, exploratory analysis, churn rate calculations, visualization, and dashboard development.

1. Data Review and Preparation

The dataset was reviewed to understand its structure, customer attributes, and churn indicator. Relevant variables were identified for analysis.

2. Data Analysis Using Microsoft Excel

Microsoft Excel was used to organize the data, calculate churn rates, compare customer segments, and prepare charts and key performance indicators.

3. Exploratory Data Analysis (EDA)

Customer churn was analysed across geography, gender, age, tenure, credit score, number of products, salary, account balance, and engagement status to identify differences between customer segments.

4. Dashboard Development Using Python and Streamlit

Python and Streamlit were used to develop an interactive dashboard containing key performance indicators, visualizations, filters, and a Customer Explorer.

5. Interpretation of Results

The results were interpreted to identify segments with higher observed churn rates and develop business recommendations based on the findings.

The analysis is descriptive: observed differences between customer segments do not, by themselves, establish the causes of customer churn.

7. EXPLORARTORY DATA ANALYSIS

7.1 Overview of Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to examine customer churn patterns and understand how churn rates vary across different customer segments. Microsoft Excel was used to calculate and compare churn rates, organize the results, and prepare visualizations.

The analysis covered geography, gender, age, tenure, credit score, number of banking products, estimated salary, account balance, and customer engagement status. The overall churn rate was also examined to establish a baseline for comparison.

The following subsections present the key results obtained from the analysis. The reported percentages represent observed churn rates within each customer segment.

7.2 Overall Customer Churn Analysis

The overall customer churn rate was analysed to understand the proportion of customers who exited the bank compared with those who remained. The dataset contains 10,000 customer records, and the overall observed churn rate was **20.37%**.

This result indicates that approximately one-fifth of the customers in the dataset had exited the bank. The overall churn rate provides a baseline for comparing churn across different customer segments and identifying groups with relatively higher observed churn rates.

The analysis establishes the starting point for further investigation into geographic, demographic, financial, and customer engagement patterns.

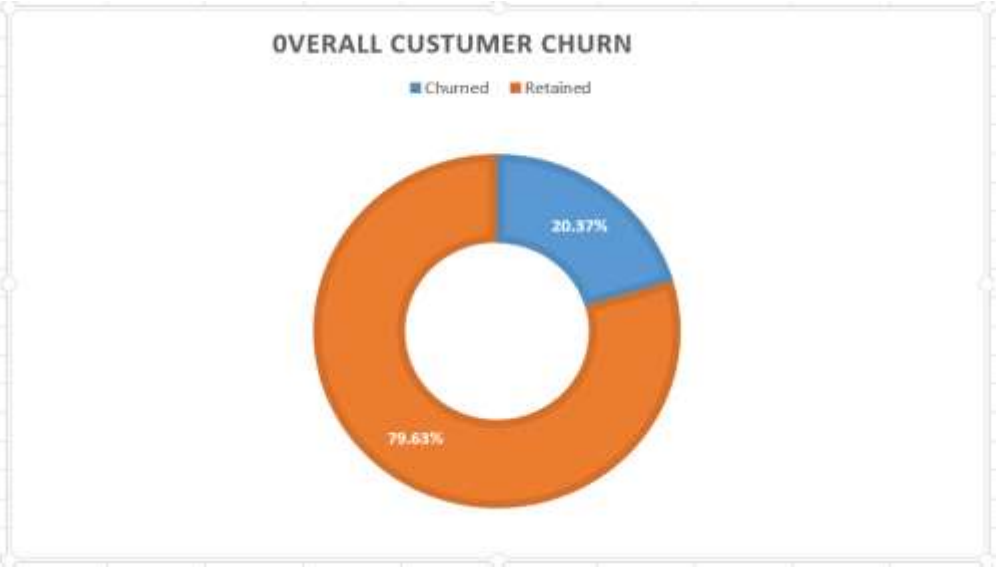


Figure: 7.2 Overall Customer Churn Rate

7.3 Geography-wise Customer Churn Analysis

Customer churn was analysed across France, Germany, and Spain to identify geographical differences in customer exit rates. The results showed that Germany had the highest observed churn rate among the three countries.

Germany recorded a churn rate of **32.44%**, while France recorded **16.15%** and Spain recorded **16.67%**. These results indicate that the proportion of customers who exited was considerably higher in Germany than in France and Spain.

The findings suggest that the German customer segment may require further investigation to understand the factors associated with its higher observed churn rate. However, the results alone do not establish the reasons for this difference.

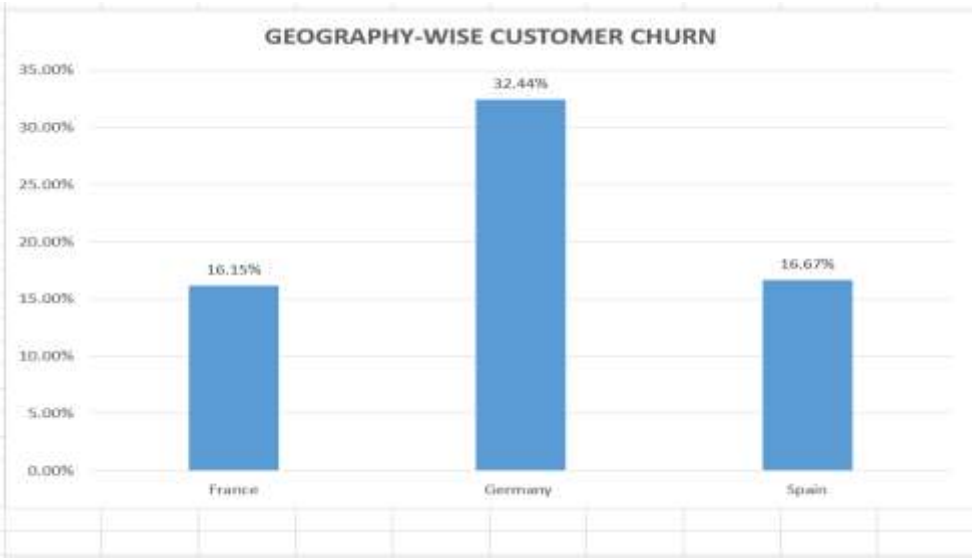


Figure: 7.3 Geography wise customer churn

7.4 Gender-wise Customer Churn Analysis

Customer churn was analysed by gender to compare the observed exit rates among female and male customers. The results showed a difference in churn rates between the two groups.

Female customers recorded an observed churn rate of **25.07%**, while male customers recorded **16.46%**. This indicates that the proportion of female customers who exited was higher than that of male customers in the dataset.

These findings highlight a difference in churn patterns across the two gender groups. Further investigation would be required to understand whether other factors, such as age, geography, product usage, or customer engagement, are associated with this difference.

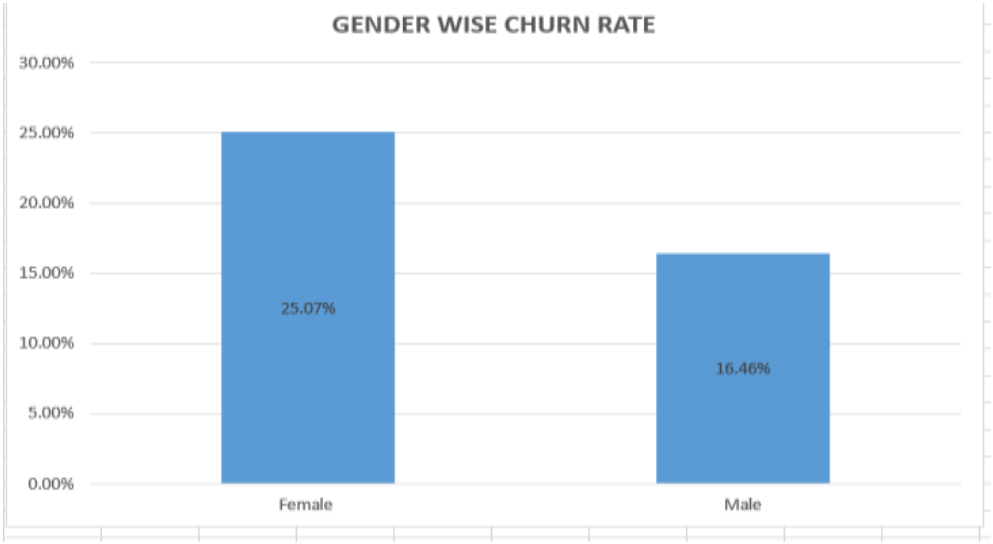


Figure: 7.4 Customer churn rate by Gender

7.5 Age-wise Customer Churn Analysis

Customer churn was analysed across different age groups to understand how observed exit rates varied with customer age. The analysis considered four age categories: below 30 years, 30–45 years, 46–60 years, and above 60 years.

The results showed that customers aged **46–60 years had the highest observed churn rate of 51.12%**. Customers below 30 years recorded a churn rate of **7.56%**, while the 30–45 age group recorded **15.30%**. Customers above 60 years had an observed churn rate of **24.78%**.

These findings indicate substantial differences in churn rates across the age groups. The 46–60 age group may therefore be a priority for further investigation into customer needs, product usage, and engagement patterns. However, the analysis does not establish that age itself causes customer churn.

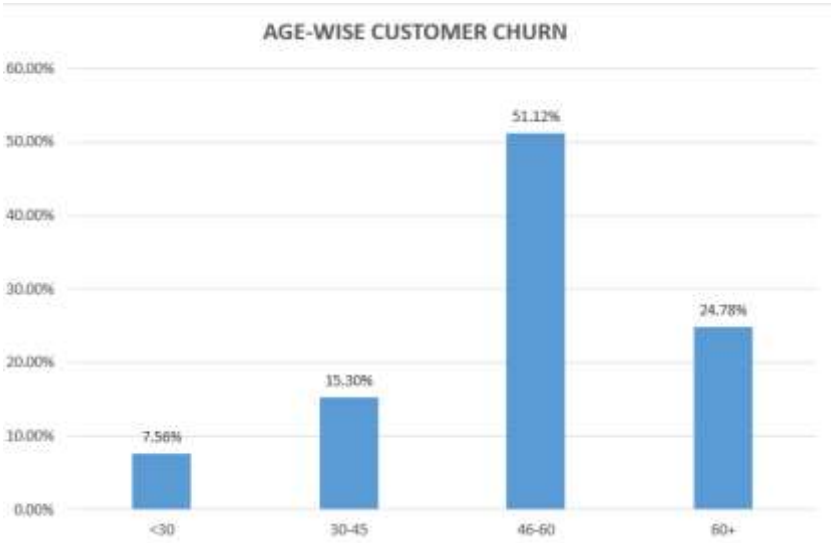


Figure: 7.5 customer churn rate by age group

7.6 Tenure-wise Customer Churn Analysis

Customer tenure refers to the length of time a customer has maintained a relationship with the bank. The analysis examined churn rates across new, mid-term, and long-term customer segments to understand whether observed churn differed according to customer tenure.

The results showed that new customers recorded an observed churn rate of **21.15%**, while both mid-term and long-term customers recorded churn rates of **19.67%**.

The difference between these rates was relatively small. New customers showed a slightly higher observed churn rate than the other two groups. This suggests that the early customer relationship may be worth examining further when developing customer engagement and retention strategies. However, the results alone do not establish why customers leave.

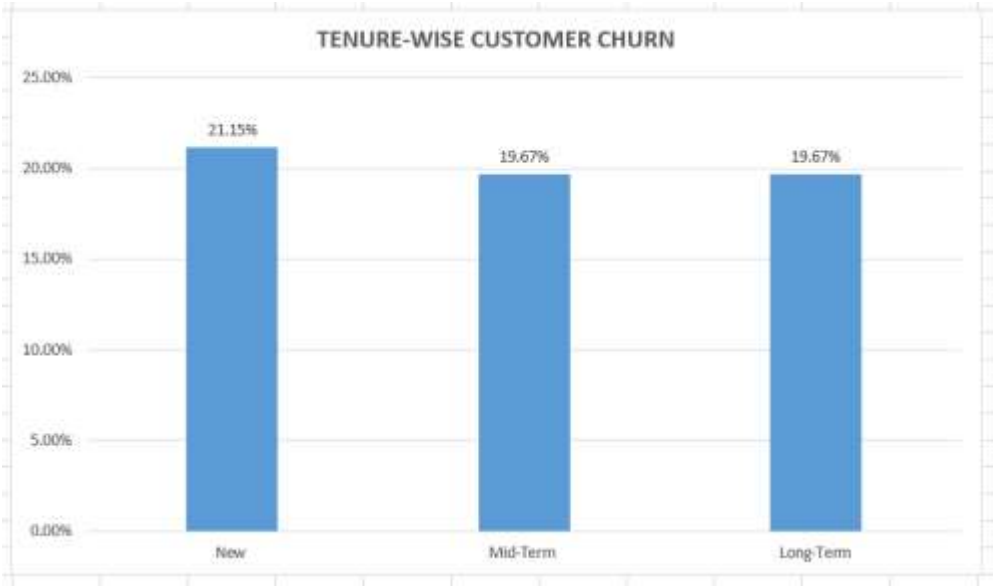


Figure: 7.6 Customer churn rate by tenure group

7.7 Credit Score Analysis

Customer churn was analysed across low, medium, and high credit score categories to examine differences in observed churn rates.

The results showed that customers in the low credit score category recorded an observed churn rate of **22.02%**. The medium credit score category recorded **20.56%**, while the high credit score category had the lowest rate at **19.32%**.

The findings show a gradual difference in observed churn rates across the three credit score categories, with the low credit score group recording a slightly higher rate than the other groups. These results can help identify segments for further analysis, but they do not establish credit score as a direct cause of customer churn.



Figure: 7.7 Customer churn rate by credit score category

7.8 Number of Products Analysis

Customer churn was analysed according to the number of banking products held by each customer. The analysis considered customers holding one, two, three, or four products to compare the observed churn rates across these groups.

The results showed that customers holding one product recorded a churn rate of **27.71%**, while customers holding two products recorded **7.58%**. Customers holding three products had a churn rate of **82.71%**, and customers holding four products recorded **100.00%**.

The results show substantial differences in observed churn rates across product categories, particularly among customers holding three or four products. These groups should be examined further to understand their customer counts and characteristics before drawing business conclusions. The high percentages alone do not establish the reasons for churn.

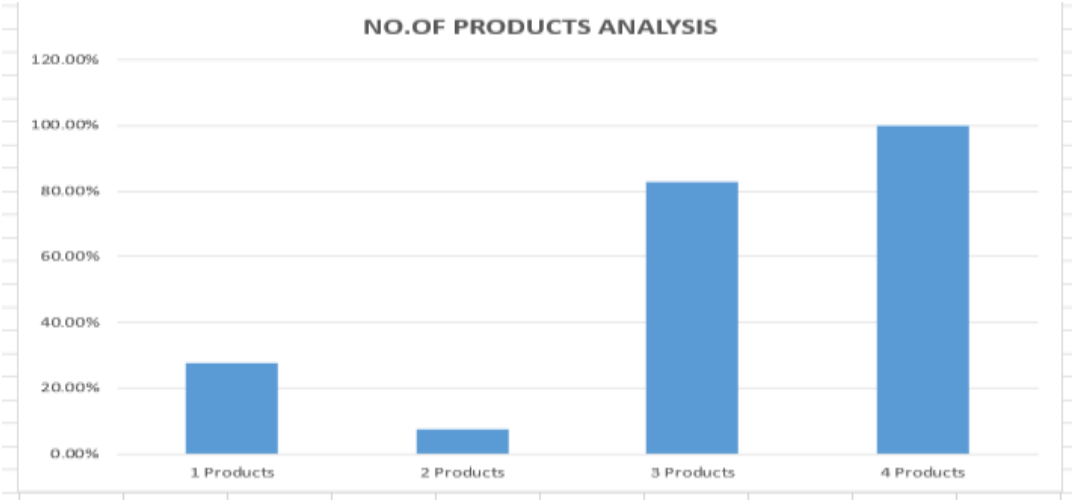


Figure: 7.8 Customer churn rate by number of products

7.9 Salary and Balance Analysis

Customer churn was analysed across estimated salary categories and account balance segments to examine differences in observed churn rates.

7.9.1 Salary Analysis

Customers in the low salary category recorded an observed churn rate of **19.93%**, while the medium salary category recorded **19.87%**. Customers in the high salary category had a slightly higher churn rate of **20.84%**. The differences between these three categories were relatively small, indicating that the observed churn rates were similar across the salary segments.

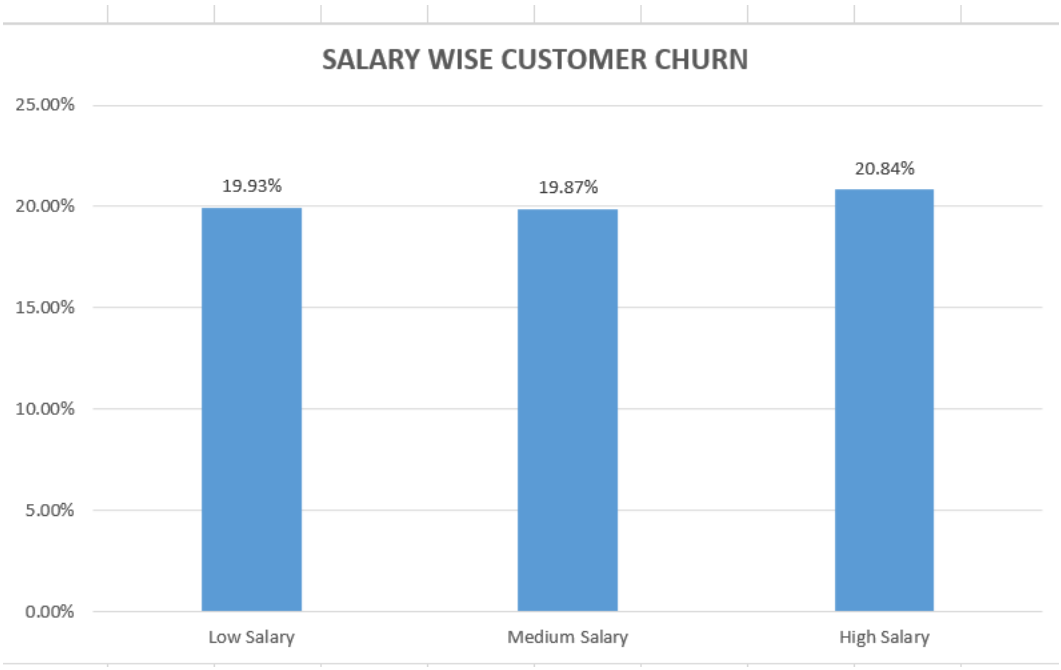


Figure: 7.9.1 customer churn rate by salary category

7.9.2

Balance Analysis

The balance analysis showed that customers with zero balances recorded an observed churn rate of **13.82%**. Customers in the low-balance segment recorded **20.82%**, while customers in the high-balance segment recorded **24.98%**.

These results show differences in observed churn rates across account balance segments, with the high-balance segment recording the highest rate. Further investigation would be needed to understand the characteristics and needs of these customers. These findings describe patterns in the dataset and do not establish that salary or account balance directly causes churn.



Figure: 7.9.2 customer churn rate by account balance segment

7.10 Customer Engagement Analysis

Customer engagement was analysed by comparing the observed churn rates of active and inactive customers. This analysis helps examine whether customer activity status is associated with differences in customer retention patterns.

The results showed that **inactive customers recorded an observed churn rate of 26.85%**, compared with **14.27% among active customers**. The difference indicates that inactive customers had a higher observed churn rate than active customers in the dataset.

This finding highlights customer engagement as an important area for further investigation. Banking institutions could examine whether suitable engagement initiatives, service follow-ups, and personalized communication help address the needs of less-active customers. However, the analysis alone does not prove that inactivity directly causes churn.

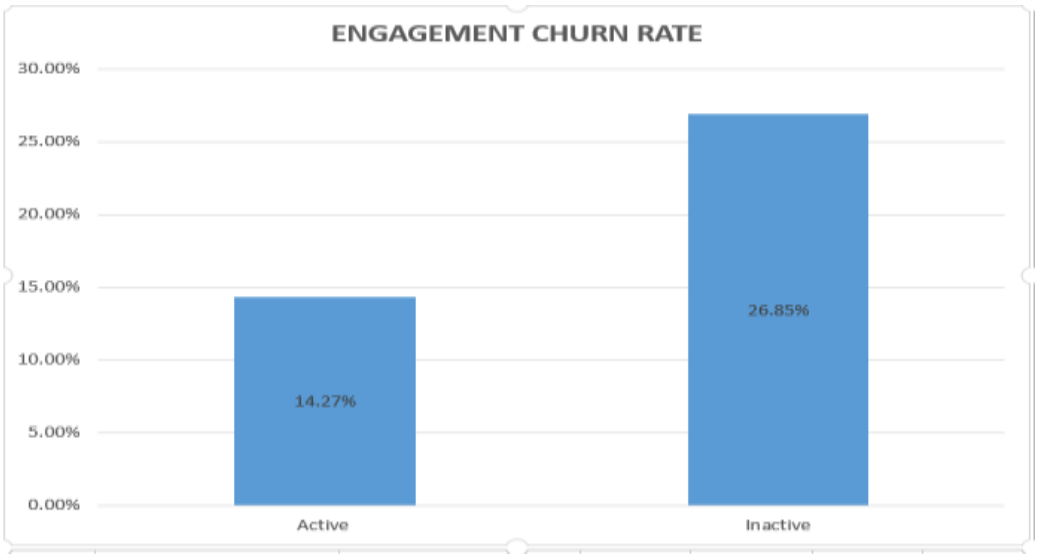


Figure: 7.10 Customer churn rate by engagement status

8.1 Key Findings

The analysis of the European banking customer dataset revealed the following findings:

1. **Overall Customer Churn:** The overall observed churn rate was 20.37%, providing a baseline for evaluating customer churn across different segments.
2. **Geographical Differences:** Germany recorded the highest observed churn rate at 32.44%, compared with 16.15% in France and 16.67% in Spain.
3. **Age Group Differences:** Customers aged 46–60 recorded an observed churn rate of 51.12%, the highest among the age groups analysed. Customers below 30 recorded the lowest rate at 7.56%.
4. **Gender Differences:** Female customers recorded an observed churn rate of 25.07%, compared with 16.46% among male customers.
5. **Customer Tenure:** New customers recorded a churn rate of 21.15%, while mid-term and long-term customers each recorded 19.67%.
6. **Credit Score:** The low credit score category recorded a churn rate of 22.02%, compared with 20.56% for medium and 19.32% for high credit scores.
7. **Number of Products:** Customers holding three products recorded a churn rate of 82.71%, while those holding four products recorded 100%. These results require further examination alongside the number of customers in each category.
8. **Salary and Account Balance:** Salary-wise churn rates were relatively similar, ranging from 19.87% to 20.84%. The high-balance segment recorded a churn rate of 24.98%, compared with 13.82% among customers with zero balances.
9. **Customer Engagement:** Inactive customers recorded a churn rate of 26.85%, compared with 14.27% among active customers.

8.2 Overall Interpretation

The findings show that observed churn rates vary across several customer segments. Geography, age group, engagement status, and number of banking products displayed notable differences in churn rates within the dataset.

These findings can help prioritize further investigation into customer segments with higher observed churn. However, the results are descriptive and do not independently establish the causes of customer churn. Customer counts and other relevant characteristics should also be considered before making business decisions.

9. BUSINESS RECOMMENDATIONS

Based on the customer churn analysis, the following recommendations are proposed:

1. **Geographical Analysis:** Investigate the reasons behind Germany's higher observed churn rate of 32.44%.
2. **Age-based Analysis:** Examine the needs and behaviour of customers aged 46–60, who recorded an observed churn rate of 51.12%.
3. **Customer Engagement:** Explore suitable engagement initiatives for inactive customers, whose observed churn rate was 26.85%.
4. **Product Analysis:** Verify customer counts and investigate the unusually high churn rates among customers holding three or four products.
5. **Customer Retention:** Examine the needs of high-balance customers, whose observed churn rate was 24.98%.
6. **Performance Monitoring:** Use the Streamlit dashboard to monitor churn indicators and identify customer segments requiring further investigation.

These recommendations provide areas for further analysis and potential retention initiatives. Their effectiveness should be evaluated using additional customer information and subsequent results.

10. STREAMLIT DASHBOARD

10.1 Dashboard Overview

An interactive dashboard was developed using Python and Streamlit to present the results of the European banking customer churn analysis. The dashboard provides a visual and interactive way to examine customer churn patterns and compare different customer segments.

10.2 Dashboard Features

The dashboard includes the following features:

1. **Key Performance Indicators (KPIs):** Displays total customers, churned customers, retained customers, and churn rate based on the selected filters.
2. **Geography-wise Analysis:** Compares customer churn rates across France, Germany, and Spain.
3. **Age-wise Analysis:** Visualizes churn patterns across customer ages.
4. **Tenure-wise Analysis:** Examines churn rates according to customer tenure.
5. **Balance and Salary Analysis:** Presents churn patterns across balance and estimated salary segments.
6. **Interactive Filters:** Allows users to filter customer records and view updated results.
7. **Customer Explorer:** Displays customer-level information for the selected segment.
8. **Key Insights:** Summarizes the selected segment's customer count, churn rate, and number of churned customers.

10.3 Dashboard Benefits

The dashboard brings important churn indicators and visualizations together in one interface. Its interactive features allow users to explore customer segments, compare results, and identify areas that may require further investigation. It demonstrates the practical application of Python and Streamlit in presenting data analysis results.



Figure: 10.3 Customer churn analysis dashboard

11. LIMITATIONS

The project has the following limitations:

1. **Dataset Scope:** The analysis is based on the available European banking customer dataset and may not represent all banking customers.
2. **Descriptive Analysis:** The project identifies patterns and differences in observed churn rates but does not establish the causes of customer churn.
3. **Limited Variables:** The analysis is restricted to the variables available in the dataset. Additional information about customer feedback, service quality, and reasons for leaving could provide deeper insights.
4. **Dashboard Scope:** The Streamlit dashboard presents the available data through interactive filters, KPIs, charts, and customer-level information. Its findings depend on the quality and scope of the underlying dataset.

12. FUTURE SCOPE

The project can be extended in several ways to improve customer churn analysis and support better business decision-making.

1. **Advanced Analytics:** Statistical analysis and machine learning models could be explored to identify factors associated with customer churn and estimate churn risk.
2. **Additional Customer Data:** Customer feedback, transaction history, service interactions, and reasons for leaving could be incorporated to gain deeper insights.
3. **Enhanced Dashboard:** Additional visualizations, more detailed filters, and improved reporting features could be developed to support further analysis.
4. **Customer Retention Strategies:** Future work could evaluate targeted retention initiatives and measure their effectiveness using subsequent customer data.
5. **Regular Monitoring:** The dashboard could be extended to analyse updated datasets periodically and monitor changes in churn patterns over time.

These improvements could help develop a more comprehensive approach to customer churn analysis and strengthen the practical value of the project.

13. EXECUTIVE SUMMARY

This project, titled **"Customer Segmentation & Churn Pattern Analytics in European Banking,"** focuses on analysing customer churn patterns using a European banking customer dataset containing 10,000 customer records. The project was completed as part of the Unified Mentor internship in the Finance Analyst domain.

The main objective was to examine customer characteristics and identify differences in observed churn rates across geographical, demographic, financial, product-related, and engagement segments.

Microsoft Excel was used to calculate churn rates, compare customer groups, and prepare analytical summaries and visualizations. Python and Streamlit were used to develop an interactive dashboard displaying key performance indicators, charts, filters, and customer-level information.

The overall observed churn rate was **20.37%**. Germany recorded a churn rate of **32.44%**, while customers aged 46–60 recorded **51.12%**. Inactive customers showed a churn rate of **26.85%**, compared with **14.27%** among active customers. Female customers recorded a churn rate of **25.07%**, compared with **16.46%** among male customers.

The analysis identified customer segments that may require further investigation. Based on these findings, recommendations were proposed to examine geographical differences, understand customer needs, investigate product-related churn patterns, and monitor customer engagement.

The Streamlit dashboard provides an interactive interface for exploring customer segments and reviewing churn indicators. Overall, the project demonstrates the practical use of data analysis and visualization to support evidence-based business analysis and customer retention planning.

14. CONCLUSION

The project titled **"Customer Segmentation & Churn Pattern Analytics in European Banking"** analysed customer churn patterns using a dataset containing 10,000 customer records. The project applied Microsoft Excel, Python, and Streamlit to examine customer segments and present the results through data analysis and visualization.

The overall observed churn rate was 20.37%. The analysis identified differences across geographical, demographic, financial, product-related, and customer engagement segments. Germany, customers aged 46–60, and inactive customers recorded notably higher observed churn rates in their respective comparisons.

An interactive Streamlit dashboard was developed to display key performance indicators, charts, filters, customer records, and segment-level insights. This dashboard makes it easier to explore the data and identify areas that may require further investigation.

The project provided practical experience in data analysis, calculating and interpreting churn rates, creating visualizations, and developing an interactive dashboard. It demonstrates how analytical tools can help organize customer information and support evidence-based business decision-making.

Overall, the project establishes a foundation for further investigation into customer retention. Additional customer information and future analysis would be required to establish the reasons for churn and evaluate the effectiveness of retention strategies.

15. REFERENCES

1. Unified Mentor. Internship project instructions and project submission guidelines.
2. Microsoft. Microsoft Excel documentation and support resources. https://support.microsoft.com/excel
3. Streamlit. Streamlit documentation. https://docs.streamlit.io/
4. Plotly. Plotly Python documentation. https://plotly.com/python/
5. Python Software Foundation. Python documentation. https://docs.python.org/3/